

Capstone Reflection Group 6: Pentachess

Team members: Brandon Bosman, Mian Rafay, Karl Sader, Saad Tariq, Emaan Khan

Github repo: <https://github.com/tariqs26/pentachess>

Achievements

Throughout this project, we all gained some form of practical experience in web development. We were able to work together as a team to apply our different strengths to different aspects of the project and overcome many challenges throughout the development stage through the use of effective problem-solving.

P0 and P1 Original Requirements 100% Completed

Priority	Requirement
P0	Ability to register for an account and log in.
P1	Ability to update login credentials and account information.
P0	Ability to start a local game.
P0	Ability to start an online game.
P0	Ability to join an online game (random search or custom invite code). (50%)
P1	Ability to forfeit once in a game.
P0	Display the board and all pieces, clearly indicating the user's color.
P0	Ability to click and drag a piece from the player's respective side upon their turn to make a legal move.
P0	Highlight all positions on the board to which the selected piece can move.
P0	Update the board state when a piece is moved to a valid spot; reset it to its original position for invalid moves.
P0	End the game on detection of checkmate, draw, no time remaining, resignation and display a message indicating the winner.
P1	Log each move with its notation and timestamp, displayed in reverse chronological order.
P0	Update the timer to count down based on the active player and sync timers after each turn.
P0	Ensure sessions expire after a period of inactivity, upon logging out, or upon forfeiting.
P0	Create endpoints for authentication.
P0	Create endpoints for updating user account information (e.g., name, username, email, password, delete account, etc.).

P0	Create endpoints for game management (e.g., game state, move history, etc.).
P0	Create a WebSocket endpoint for joining an online game.
P0	Create a WebSocket endpoint for leaving an online game.
P0	Create a WebSocket endpoint for move events, which ensures the game state is in sync between the two clients.

Incomplete Requirements

Priority	Requirement	Reason
P0	Option for password recovery. (0%)	Required a third party email service, however we wanted to keep costs to a minimum.
P0	Ability to set a custom timer for local and online games, and game code and password for online games. (50%)	Too difficult given time constraints.
P1	Responsive user interface that adjusts to any device screen size. (90%)	N/A (above 80%)
P1	Display all captured pieces for both players and the respective differences in their total values. (90%)	N/A (above 80%)

What Worked and What Did Not

One of the most important things that worked in our project was our weekly meetings. Every week, we got together to discuss completed tasks and future tasks that remained to be done, which helped us stay on track and gave us a clear view of what everyone was currently working on. Toward the end of the project, what was tricky was time management, as our original timeline for the project wasn't 100% fleshed out. Nonetheless, we finished almost everything necessary for the project to be usable.

What We Learned

Our team has gained valuable insights into both the technical aspects of development and the essential skills required for successful project management. Throughout this project, we enhanced our time management abilities, as we had to meet tight deadlines while keeping up with the course load from other courses. With continuous updates from Dr. Paul regarding the project, we had to ensure that we set up an environment to take advantage of continuous integration and delivery. Not only that, but visual demos would also help Dr. Paul test our current version to ensure small bugs are corrected as we continue to develop. Teamwork played a critical role, as many of the weekly tasks depended on each other, requiring us to collaborate closely and efficiently. Through weekly team meetings, our communication skills improved greatly, as we were able to effectively convey our ideas clearly and maintain regular updates to ensure the project's progress aligned with our set goals. In addition, we developed strong problem-solving skills along with adaptability. Addressing challenges related to game mechanics and functionality pushed us to expand our knowledge and quickly acquire new skills to meet the project's demands. Overall, this experience provided us with a strong overview of what core skills are needed to become a successful developer working with a team to accomplish a common goal.